

Symbol-LLM

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Introduction to Symbol-LLM

- A series of models made to create a symbol oriented interface for LLMs
- LLMs are probabilistic models that follow patterns
- LLMs struggle with complex logic but are good at understanding language
- Deterministic symbolic engines draw conclusions from facts and reason
- Symbol-LLMs are LLMs combined with symbolic engines
 - Take the understanding from LLMs and the faithful reasoning from the symbolic engines

Limitations of LLMs

- Rely solely on natural language
 - Can only be applicable to facts and common knowledge
- First-order logic is better represented in symbolic forms rather than natural language
- Symbol forms are more concise and accurate involving robot interaction of calling extra tools via programming language

Symbolic Data Collection

- Researchers curated a data collection consisting of 34 tasks and around 20 distinct symbolic families
- First to treat symbolic data in unified view to explore connections among different symbols
- Covers many domains
 - Math reasoning
 - Symbolic reasoning
 - Logical reasoning
 - Robotic planning
 - Visual reasoning

Data Sources

- Around 88.3% of symbolic data collected from existed benchmarks
- Around 5.8% of the data was generated by prompting powerful models such as GPT-4 to create text-to symbol pairs
- Rest of the data was generated using Symbol-evol strategy to prevent model memorization

Symbol-evol Strategy

- Replaces original symbolic definitions with randomly generating strings
- Enhances the diversity of symbolic systems
- Forces the model to learn to follow instructions rather than memorizing specific symbolic patterns
- Diverse symbol instruction samples are from existing tasks

Tuning

- Injection stage: focuses on balancing the model's capabilities in symbolic and natural language tasks
 - Prioritizes the exploitation of the connection between different symbols
 - Enables the model to thoroughly learn a wide range of symbolic knowledge
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